How to Simulate Like a Quant Desk. Every Model, Every Formula, Runnable Code

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This isn't a list of techniques.

It's a story - one that starts with a coin flip and ends with institutional-grade simulation engines.

Each section builds on the last. Skip ahead and the math won't make sense. Read it in order and by the end you'll have runnable code for every layer of the stack.

Disclaimer:
Not Financial Advice & Do Your Own Research

Part I: The Coin Flip That Breaks Everything

You're staring at a Polymarket contract. "Will the Fed cut rates in March?" YES is trading at $0.62.

Your instinct says: that's a 62% probability. Maybe you think it should be 70%. So you buy.

Congratulations. You just did what every retail trader does. You treated a prediction market contract like a coin flip with a known bias, estimated your own bias, and bet the difference.

> You have no idea how confident to be in your 70% estimate.

> You don't know how it should change when tomorrow's jobs report drops.
>
> You don't know how it correlates with the six other Fed-related contracts on Polymarket.
>
> You don't know whether the price path between now and resolution will let you exit at a profit even if you're eventually right.

A coin flip has one parameter: p.

A prediction market contract embedded in a portfolio of correlated events, with time-varying information flow, order book dynamics, and execution risk, has dozens.

Part II: Monte Carlo. The Foundation Nobody Respects Enough

Every simulation in this article ultimately reduces to Monte Carlo: draw samples from a distribution, compute a statistic, repeat.

The estimator for event probability $p=P(A)$ is just the sample mean:

The Central Limit Theorem gives you the convergence rate: $O(N^{-1/2}$, with variance $Var(\hat{p}_N)=p(1-p)/N$.

T**he variance is maximized at p=0.5p** A contract trading at 50 cents the most uncertain, most actively traded contract on the platform is exactly where your Monte Carlo estimates are least precise.

To hit ±0.01 precision at 95% confidence when p=0.50:

That's manageable. But it gets worse fast when you need to simulate *paths*, not just endpoints.

Your First Runnable Simulation

Goal: Estimate the probability that an asset-linked binary contract pays off (e.g., "Will AAPL close above $200 by March 15?")

```python
import numpy as np

def simulate_binary_contract(S0, K, mu, sigma, T, N_paths=100_000):
    """
    Monte Carlo simulation for a binary contract.

    S0:    Current asset price
    K:     Strike / threshold
    mu:    Annual drift
    sigma: Annual volatility
    T:     Time to expiry in years
    N_paths: Number of simulated paths
    """
    # Simulate terminal prices via GBM
    Z = np.random.standard_normal(N_paths)
    S_T = S0 * np.exp((mu - 0.5 * sigma**2) * T + sigma * np.sqrt(T) * Z)

    # Binary payoff
    payoffs = (S_T > K).astype(float)

    # Estimate and confidence interval
    p_hat = payoffs.mean()
    se = np.sqrt(p_hat * (1 - p_hat) / N_paths)
    ci_lower = p_hat - 1.96 * se
    ci_upper = p_hat + 1.96 * se

    return {
        'probability': p_hat,
        'std_error': se,
        'ci_95': (ci_lower, ci_upper),
        'N_paths': N_paths
    }

# Example: AAPL at $195, strike $200, 20% vol, 30 days
result = simulate_binary_contract(S0=195, K=200, mu=0.08, sigma=0.20, T=30/36!
print(f"P(AAPL > $200) ≈ {result['probability']:.4f}")
print(f"95% CI: ({result['ci_95'][0]:.4f}, {result['ci_95'][1]:.4f})")
```

This works. For one contract, with one underlying, assuming lognormal dynamics. Real prediction markets break every one of those assumptions.

Evaluating Your Simulation

Before we improve the simulation, we need a way to measure how good it is. The **Brier Score** is the standard calibration metric:

```python
def brier_score(predictions, outcomes):
    """Evaluate simulation calibration."""
    return np.mean((np.array(predictions) - np.array(outcomes))**2)

# Compare two models
model_A_preds = [0.7, 0.3, 0.9, 0.1]  # sharp, confident
model_B_preds = [0.5, 0.5, 0.5, 0.5]  # always uncertain
actual_outcomes = [1, 0, 1, 0]

print(f"Model A Brier: {brier_score(model_A_preds, actual_outcomes):.4f}")  #
print(f"Model B Brier: {brier_score(model_B_preds, actual_outcomes):.4f}")  #
```

A Brier score below 0.20 is good.

Below 0.10 is excellent.

The best election forecasters (538, Economist) historically achieve 0.06-0.12 on presidential races.

If your simulation can beat that, you have edge.

Part III: When 100,000 Samples Aren't Enough

Now the story escalates.

Polymarket hosts contracts on extreme events. "Will the S&P 500 drop 20% in one week?" is trading at $0.003. With crude Monte Carlo at 100,000 samples, you might see zero or one hit.

Your estimate is either 0.00000 or 0.00001 - both useless.

This isn't a theoretical problem. It's the reason most retail traders can't properly evaluate tail-risk contracts.

Make Rare Events Common

Importance sampling replaces the original probability measure with one that oversamples the rare region, then corrects the bias with a likelihood

L**ikelihood ratio** or Radon-Nikodym derivative

Not useful directly, but it tells you what to aim for.

The practical workhorse is **exponential tilting**.

If your underlying follows a random walk with increments Δ_ihaving moment generating function M(γ)=E[e^γΔ], you tilt the distribution:

choosing γ to make the rare event typical. For a contract that pays off when a sum exceeds a large threshold, γ solves the **Lundberg equation** M(γ)=1.

Importance Sampling for Tail-Risk Contracts

```python
def rare_event_IS(S0, K_crash, sigma, T, N_paths=100_000):
    """
    Importance sampling for extreme downside binary contracts.

    Example: P(S&P drops 20% in one week)
    """
    K = S0 * (1 - K_crash)  # e.g., 20% crash threshold

    # Original drift (risk-neutral)
    mu_original = -0.5 * sigma**2

    # Tilted drift: shift the mean toward the crash region
    # Choose mu_tilt so the crash threshold is ~1 std dev away instead of ~4
    log_threshold = np.log(K / S0)
    mu_tilt = log_threshold / T  # center the distribution on the crash

    Z = np.random.standard_normal(N_paths)

    # Simulate under TILTED measure
    log_returns_tilted = mu_tilt * T + sigma * np.sqrt(T) * Z
    S_T_tilted = S0 * np.exp(log_returns_tilted)

    # Likelihood ratio: original density / tilted density
    log_returns_original = mu_original * T + sigma * np.sqrt(T) * Z
    log_LR = (
        -0.5 * ((log_returns_tilted - mu_original * T) / (sigma * np.sqrt(T))
```

```
        + 0.5 * ((log_returns_tilted - mu_tilt * T) / (sigma * np.sqrt(T)))**2
    )
    LR = np.exp(log_LR)

    # IS estimator
    payoffs = (S_T_tilted < K).astype(float)
    is_estimates = payoffs * LR

    p_IS = is_estimates.mean()
    se_IS = is_estimates.std() / np.sqrt(N_paths)

    # Compare with crude MC
    Z_crude = np.random.standard_normal(N_paths)
    S_T_crude = S0 * np.exp(mu_original * T + sigma * np.sqrt(T) * Z_crude)
    p_crude = (S_T_crude < K).mean()
    se_crude = np.sqrt(p_crude * (1 - p_crude) / N_paths) if p_crude > 0 else

    return {
        'p_IS': p_IS, 'se_IS': se_IS,
        'p_crude': p_crude, 'se_crude': se_crude,
        'variance_reduction': (se_crude / se_IS)**2 if se_IS > 0 else float('
    }

result = rare_event_IS(S0=5000, K_crash=0.20, sigma=0.15, T=5/252)
print(f"IS estimate:    {result['p_IS']:.6f} ± {result['se_IS']:.6f}")
print(f"Crude estimate: {result['p_crude']:.6f} ± {result['se_crude']:.6f}")
print(f"Variance reduction factor: {result['variance_reduction']:.1f}x")
```

On extreme contracts, IS can reduce variance by **factors of 100–10,000x**.

This means 100 IS samples give better precision than 1,000,000 crude samples.

That's not a marginal improvement. It's the difference between "we can't price this" and "we're trading it."

Part IV: Sequential Monte Carlo for Real-Time Updating

But what I need to do while the story shifts from static estimation to dynamic simulation?

Imagine:
It's election night. 8:01 PM EST. Florida polls just closed. Early returns show a 3-point shift toward one candidate.

Your model needs to update *instantly* incorporating this new data point into the probability estimate for not just Florida, but Ohio, Pennsylvania, Michigan, and every correlated state.

This is the **filtering problem**, and the tool is **Sequential Monte Carlo** particle filters.

The State-Space Model

Define:

> **Hidden state** x_t: the "true" probability of the event (unobserved)
>
> **Observation** y_t: market prices, poll results, vote counts, news signals

The state evolves via a **logit random walk** (keeps probabilities bounded):

Observations are noisy readings of the true state:

The Bootstrap Particle Filter

The algorithm maintains N "particles" - each one a hypothesis about the true probability and reweights them as data arrives:

```
1. INITIALIZE: Draw x_0^{(i)} ~ Prior  for i = 1,...,N
   Set weights w_0^{(i)} = 1/N

2. FOR each new observation y_t:
   a. PROPAGATE:  x_t^{(i)} ~ f( · | x_{t-1}^{(i)} )
   b. REWEIGHT:   w_t^{(i)} ∝ g( y_t | x_t^{(i)} )
   c. NORMALIZE:  w̃_t^{(i)} = w_t^{(i)} / Σ_j w_t^{(j)}
   d. RESAMPLE if ESS = 1/Σ(w̃_t^{(i)})² < N/2
```

Particle Filter for a Live Prediction Market

```python
import numpy as np
from scipy.special import expit, logit  # sigmoid and logit

class PredictionMarketParticleFilter:
    """
    Sequential Monte Carlo filter for real-time event probability estimation.

    Usage during a live event (e.g., election night):
        pf = PredictionMarketParticleFilter(prior_prob=0.50)
        pf.update(observed_price=0.55)   # market moves on early returns
        pf.update(observed_price=0.62)   # more data
        pf.update(observed_price=0.58)   # partial correction
        print(pf.estimate())             # filtered probability
    """
    def __init__(self, N_particles=5000, prior_prob=0.5,
                 process_vol=0.05, obs_noise=0.03):
        self.N = N_particles
        self.process_vol = process_vol
        self.obs_noise = obs_noise

        # Initialize particles around prior
        logit_prior = logit(prior_prob)
        self.logit_particles = logit_prior + np.random.normal(0, 0.5, N_parti
        self.weights = np.ones(N_particles) / N_particles
        self.history = []

    def update(self, observed_price):
```

```
        """Incorporate a new observation (market price, poll result, etc.)"""
        # 1. Propagate: random walk in logit space
        noise = np.random.normal(0, self.process_vol, self.N)
        self.logit_particles += noise

        # 2. Convert to probability space
        prob_particles = expit(self.logit_particles)

        # 3. Reweight: likelihood of observation given each particle
        log_likelihood = -0.5 * ((observed_price - prob_particles) / self.obs_
        log_weights = np.log(self.weights + 1e-300) + log_likelihood

        # Normalize in log space for stability
        log_weights -= log_weights.max()
        self.weights = np.exp(log_weights)
        self.weights /= self.weights.sum()

        # 4. Check ESS and resample if needed
        ess = 1.0 / np.sum(self.weights**2)
        if ess < self.N / 2:
            self._systematic_resample()

        self.history.append(self.estimate())

    def _systematic_resample(self):
        """Systematic resampling - lower variance than multinomial."""
        cumsum = np.cumsum(self.weights)
        u = (np.arange(self.N) + np.random.uniform()) / self.N
        indices = np.searchsorted(cumsum, u)
        self.logit_particles = self.logit_particles[indices]
        self.weights = np.ones(self.N) / self.N

    def estimate(self):
        """Weighted mean probability estimate."""
        probs = expit(self.logit_particles)
        return np.average(probs, weights=self.weights)

    def credible_interval(self, alpha=0.05):
        """Weighted quantile-based credible interval."""
        probs = expit(self.logit_particles)
        sorted_idx = np.argsort(probs)
        sorted_probs = probs[sorted_idx]
        sorted_weights = self.weights[sorted_idx]
        cumw = np.cumsum(sorted_weights)
        lower = sorted_probs[np.searchsorted(cumw, alpha/2)]
        upper = sorted_probs[np.searchsorted(cumw, 1 - alpha/2)]
        return lower, upper

# --- Simulate election night ---
pf = PredictionMarketParticleFilter(prior_prob=0.50, process_vol=0.03)
```

```
# Incoming observations (market prices as new data arrives)
observations = [0.50, 0.52, 0.55, 0.58, 0.61, 0.63, 0.60,
                0.65, 0.70, 0.75, 0.80, 0.85, 0.90, 0.95]

print("Election Night Tracker:")
print(f"{'Time':>6}  {'Observed':>10}  {'Filtered':>10}  {'95% CI':>20}")
print("-" * 52)

for t, obs in enumerate(observations):
    pf.update(obs)
    ci = pf.credible_interval()
    print(f"{t:>5}h  {obs:>10.3f}  {pf.estimate():>10.3f}  ({ci[0]:.3f}, {ci[1
```

Why is this better than just using the market price directly?

Because the particle filter **smooths noise and propagates uncertainty**.

When the market spikes from $0.58 to $0.65 on a single trade, the filter recognizes that the true probability might not have changed that much it tempers the update based on how volatile the observation process has been.

Part V: Three Variance Reduction Tricks That Stack

Before we leave Monte Carlo territory, here are three techniques that **combine multiplicatively** with everything above.

Free Symmetry

When the payoff function is monotone (which binary contracts always are higher prices mean higher probability of exceeding the strike), the variance reduction is guaranteed:

Typical reduction is around 50-75%. Zero extra computational cost beyond doubling the function evaluations (which you were going to do anyway).

Exploit What You Already Know

If you're simulating a binary contract {S_T > K} under stochastic volatility (no closed form), use the Black-Scholes digital price p_{BS} (which has a closed form) as a **control variate**:

Divide and Conquer

Partition the probability space into JJ J strata, sample within each, combine. The variance is always ≤ crude MC (by the law of total variance), with maximum gain from **Neyman allocation**: nj∝ωjσj (oversample strata with high variance).

```python
def stratified_binary_mc(S0, K, sigma, T, J=10, N_total=100_000):
    """
    Stratified MC for binary contract pricing.
    Strata defined by quantiles of the terminal price distribution.
    """
    n_per_stratum = N_total // J
    estimates = []

    for j in range(J):
        # Uniform draws within stratum [j/J, (j+1)/J]
```

```
        U = np.random.uniform(j/J, (j+1)/J, n_per_stratum)
        Z = norm.ppf(U)
        S_T = S0 * np.exp((-0.5*sigma**2)*T + sigma*np.sqrt(T)*Z)
        stratum_mean = (S_T > K).mean()
        estimates.append(stratum_mean)

    # Each stratum has weight 1/J
    p_stratified = np.mean(estimates)
    se_stratified = np.std(estimates) / np.sqrt(J)

    return p_stratified, se_stratified

p, se = stratified_binary_mc(S0=100, K=105, sigma=0.20, T=30/365)
print(f"Stratified estimate: {p:.6f} ± {se:.6f}")
```

Stack all three

Antithetic variates inside each stratum, with a control variate correction and you routinely achieve **100–500x** variance reduction over crude MC. This is not optional in production. This is table stakes.

Part VI: Modeling What Correlation Matrices Can't

The hierarchical Bayesian model implicitly encodes correlation through the shared national swing parameter.

But what about **tail dependence** - the tendency for extreme co-movements that don't show up in linear correlation?

In 2008, the Gaussian copula's failure to model tail dependence contributed to the global financial crisis. In prediction markets, the same issue arises: when one swing state has a surprise result, the probability that *all* swing states flip together is much higher than a Gaussian copula would predict.

Sklar's Theorem

where C is the copula (the pure dependency structure) and F_i are the marginal CDFs. You can model each market's marginal behavior separately, then glue them together with a copula that captures the dependency including in the tails.

The Tail Dependence Problem

Gaussian copula: Tail dependence $\lambda_U=\lambda_L=0$. Extreme co-movements are modeled as having zero probability.

This is **catastrophically wrong** for correlated prediction markets.

Student-t copula

With $\nu=4$ and $\rho=0.6$, tail dependence is approximately 0.18 -z an 18% probability that extreme co-movement occurs given one contract hits an extreme. Gaussian would say 0%.

Clayton copula: Lower tail dependence only ($\lambda_L=2^{-1/\theta}$. When one prediction market crashes, others follow. No upper tail dependence.

Gumbel copula: Upper tail dependence only ($\lambda_U=2-2^{1/\theta}$). Correlated positive resolutions.

Simulating Correlated Prediction Market Outcomes

python

```python
import numpy as np
from scipy.stats import norm, t as t_dist

def simulate_correlated_outcomes_gaussian(probs, corr_matrix, N=100_000):
    """Gaussian copula no tail dependence."""
    d = len(probs)
    L = np.linalg.cholesky(corr_matrix)
    Z = np.random.standard_normal((N, d))
    X = Z @ L.T
    U = norm.cdf(X)
    outcomes = (U < np.array(probs)).astype(int)
    return outcomes

def simulate_correlated_outcomes_t(probs, corr_matrix, nu=4, N=100_000):
    """Student-t copula symmetric tail dependence."""
    d = len(probs)
    L = np.linalg.cholesky(corr_matrix)
    Z = np.random.standard_normal((N, d))
    X = Z @ L.T

    # Divide by sqrt(chi-squared / nu) to get t-distributed
    S = np.random.chisquare(nu, N) / nu
    T = X / np.sqrt(S[:, None])
    U = t_dist.cdf(T, nu)
    outcomes = (U < np.array(probs)).astype(int)
    return outcomes

def simulate_correlated_outcomes_clayton(probs, theta=2.0, N=100_000):
    """Clayton copula (bivariate) lower tail dependence."""
    # Marshall-Olkin algorithm
    V = np.random.gamma(1/theta, 1, N)
    E = np.random.exponential(1, (N, len(probs)))
    U = (1 + E / V[:, None])**(-1/theta)
    outcomes = (U < np.array(probs)).astype(int)
    return outcomes


# --- Compare tail behavior ---
probs = [0.52, 0.53, 0.51, 0.48, 0.50]  # 5 swing state probabilities
state_names = ['PA', 'MI', 'WI', 'GA', 'AZ']

corr = np.array([
    [1.0, 0.7, 0.7, 0.4, 0.3],
    [0.7, 1.0, 0.8, 0.3, 0.3],
    [0.7, 0.8, 1.0, 0.3, 0.3],
```

```
    [0.4, 0.3, 0.3, 1.0, 0.5],
    [0.3, 0.3, 0.3, 0.5, 1.0],
])

N = 500_000

gauss_outcomes = simulate_correlated_outcomes_gaussian(probs, corr, N)
t_outcomes = simulate_correlated_outcomes_t(probs, corr, nu=4, N=N)

# P(sweep all 5 states)
p_sweep_gauss = gauss_outcomes.all(axis=1).mean()
p_sweep_t = t_outcomes.all(axis=1).mean()

# P(lose all 5 states)
p_lose_gauss = (1 - gauss_outcomes).all(axis=1).mean()
p_lose_t = (1 - t_outcomes).all(axis=1).mean()

# If independent
p_sweep_indep = np.prod(probs)
p_lose_indep = np.prod([1-p for p in probs])

print("Joint Outcome Probabilities:")
print(f"{'':>25}  {'Independent':>12}  {'Gaussian':>12}  {'t-copula':>12}")
print(f"{'P(sweep all 5)':>25}  {p_sweep_indep:>12.4f}  {p_sweep_gauss:>12.4f
print(f"{'P(lose all 5)':>25}  {p_lose_indep:>12.4f}  {p_lose_gauss:>12.4f}
print(f"\nt-copula increases sweep probability by {p_sweep_t/p_sweep_gauss:.1
```

This is the **exact reason** the Gaussian copula failed in 2008 and would fail again for prediction market portfolios.

The t-copula with v = 4 routinely shows 2–5x higher probability of extreme joint outcomes.

If you're trading correlated prediction market contracts without modeling tail dependence, you're running a portfolio that will blow up in exactly the scenarios that matter most.

Vine Copula

For d>5 contracts, bivariate copulas are insufficient. **Vine copulas** decompose the dd d-dimensional dependency into d(d−1)/2 bivariate conditional copulas arranged in a tree structure:

C-vine (star): One central event drives everything (e.g., presidential winner -> all policy markets)

D-vine (path): Sequential dependencies (e.g., primary results flow into general election)

R-vine (general graph): Maximum flexibility

build maximum spanning trees ordered by |τKendall|, select pair-copula families via AIC, estimate sequentially. Implementations: pyvinecopulib (Python), VineCopula (R).

Part VII: Agent-Based Simulation

Everything so far assumes you know the data-generating process and just need to simulate it.

But prediction markets are populated by *heterogeneous agents* - informed traders, noise traders, market makers, and bots whose interactions produce emergent dynamics that no closed-form SDE can capture.

The Zero-Intelligence Revelation

Markets can be efficient even when every single trader is completely irrational.

Gode & Sunder (1993) showed that zero-intelligence agents - traders who submit random orders subject only to budget constraints achieve near-100% allocative efficiency in a continuous double auction.

Farmer, Patelli & Zovko (2005) extended this to limit order books.

This explained 96% of cross-sectional spread variation on the London Stock Exchange. One parameter. 96%.

Agent-Based Prediction Market Simulator

```python
import numpy as np
from collections import deque

class PredictionMarketABM:
    """
    Agent-based model of a prediction market order book.

    Agent types:
    - Informed: know the true probability, trade toward it
    - Noise: random trades
    - Market maker: provides liquidity around current price
    """
    def __init__(self, true_prob, n_informed=10, n_noise=50, n_mm=5):
        self.true_prob = true_prob
        self.price = 0.50  # initial price
        self.price_history = [self.price]

        # Order book (simplified as bid/ask queues)
        self.best_bid = 0.49
        self.best_ask = 0.51

        # Agent populations
        self.n_informed = n_informed
        self.n_noise = n_noise
        self.n_mm = n_mm

        # Track metrics
        self.volume = 0
        self.informed_pnl = 0
        self.noise_pnl = 0

    def step(self):
        """One time step: randomly select an agent to trade."""
        total = self.n_informed + self.n_noise + self.n_mm
        r = np.random.random()

        if r < self.n_informed / total:
            self._informed_trade()
        elif r < (self.n_informed + self.n_noise) / total:
            self._noise_trade()
        else:
            self._mm_update()

        self.price_history.append(self.price)
```

```
    def _informed_trade(self):
        """Informed trader: buy if price < true_prob, sell otherwise."""
        signal = self.true_prob + np.random.normal(0, 0.02)  # noisy signal

        if signal > self.best_ask + 0.01:  # buy
            size = min(0.1, abs(signal - self.price) * 2)
            self.price += size * self._kyle_lambda()
            self.volume += size
            self.informed_pnl += (self.true_prob - self.best_ask) * size
        elif signal < self.best_bid - 0.01:  # sell
            size = min(0.1, abs(self.price - signal) * 2)
            self.price -= size * self._kyle_lambda()
            self.volume += size
            self.informed_pnl += (self.best_bid - self.true_prob) * size

        self.price = np.clip(self.price, 0.01, 0.99)
        self._update_book()

    def _noise_trade(self):
        """Noise trader: random buy/sell."""
        direction = np.random.choice([-1, 1])
        size = np.random.exponential(0.02)
        self.price += direction * size * self._kyle_lambda()
        self.price = np.clip(self.price, 0.01, 0.99)
        self.volume += size
        self.noise_pnl -= abs(self.price - self.true_prob) * size * 0.5
        self._update_book()

    def _mm_update(self):
        """Market maker: tighten spread toward current price."""
        spread = max(0.02, 0.05 * (1 - self.volume / 100))
        self.best_bid = self.price - spread / 2
        self.best_ask = self.price + spread / 2

    def _kyle_lambda(self):
        """Price impact parameter."""
        sigma_v = abs(self.true_prob - self.price) + 0.05
        sigma_u = 0.1 * np.sqrt(self.n_noise)
        return sigma_v / (2 * sigma_u)

    def _update_book(self):
        spread = self.best_ask - self.best_bid
        self.best_bid = self.price - spread / 2
        self.best_ask = self.price + spread / 2

    def run(self, n_steps=1000):
        for _ in range(n_steps):
            self.step()
        return np.array(self.price_history)
```

```
# --- Simulation ---
np.random.seed(42)

# Scenario: true probability is 0.65, market starts at 0.50
sim = PredictionMarketABM(true_prob=0.65, n_informed=10, n_noise=50, n_mm=5)
prices = sim.run(n_steps=2000)

print("Agent-Based Prediction Market Simulation")
print(f"True probability:   {sim.true_prob:.2f}")
print(f"Starting price:     0.50")
print(f"Final price:        {prices[-1]:.4f}")
print(f"Price at t=500:     {prices[500]:.4f}")
print(f"Price at t=1000:    {prices[1000]:.4f}")
print(f"Total volume:       {sim.volume:.1f}")
print(f"Informed P&L:       ${sim.informed_pnl:.2f}")
print(f"Noise trader P&L:   ${sim.noise_pnl:.2f}")
print(f"Convergence error:  {abs(prices[-1] - sim.true_prob):.4f}")
```

H**ow fast prices converge depends on the ratio of informed to noise traders**, how market maker spread responds to information flow, and why the informed traders extract profit at noise traders expense.

Part VIII: The Production Stack

Here's the complete system, from market data to trade execution:

```
LAYER 1: DATA INGESTION
- WebSocket feed from Polymarket CLOB API (real-time prices, volumes)
- News/poll feeds (NLP-processed into probability signals)
- On-chain event data (Polygon)

LAYER 2: PROBABILITY ENGINE
- Hierarchical Bayesian model (Stan/PyMC) state-level posteriors
- Particle filter real-time updating on new observations
- Jump-diffusion SDE path simulation for risk management
- Ensemble: weighted average of model outputs

LAYER 3: DEPENDENCY MODELING
- Vine copula pairwise dependencies between contracts
```

- Factor model shared national/global risk factors
- Tail dependence estimation via t-copula

LAYER 4: RISK MANAGEMENT
- EVT-based VaR and Expected Shortfall
- Reverse stress testing identify worst-case scenarios
- Correlation stress what if state correlations spike?
- Liquidity risk order book depth monitoring

LAYER 5: MONITORING
- Brier score tracking (are we calibrated?)
- P&L attribution (which model component added value?)
- Drawdown alerts
- Model drift detection

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